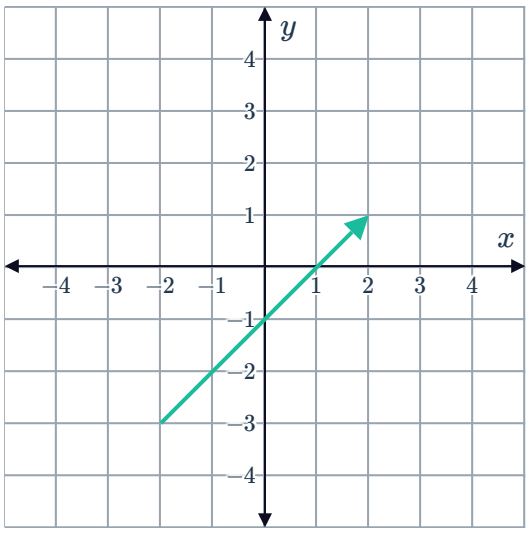
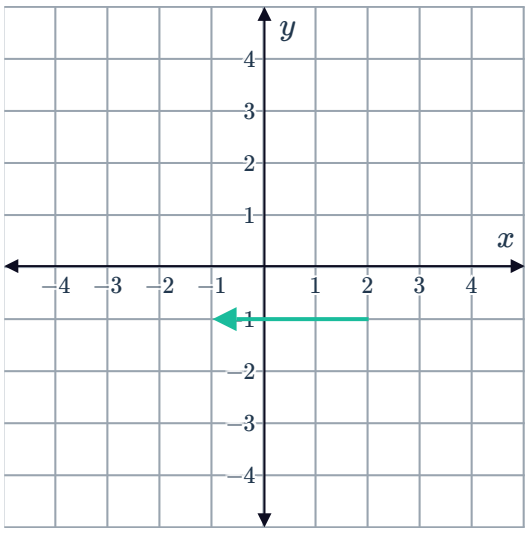
# Column vectors

**1** Write the vector represented on the following planes as a column vector:

**a**

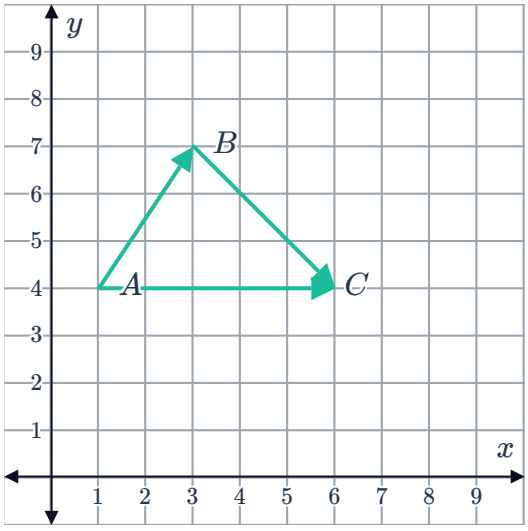



**b**




**2** Consider the following figure and write the column vector for each of the following:

**a** $\overrightarrow{AB}$ **b** $\overrightarrow{BC}$ **c** $\overrightarrow{AC}$



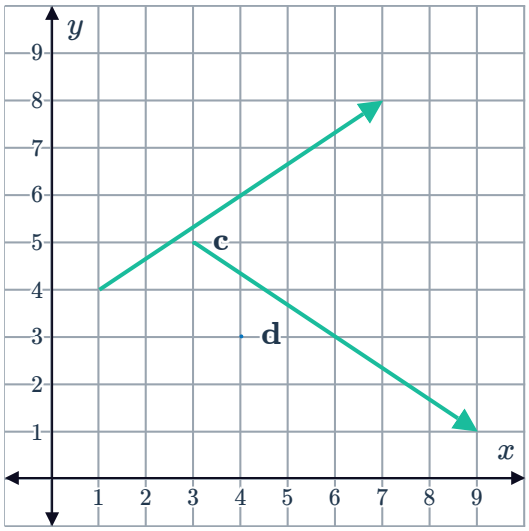

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**3** For each of the following graphs:

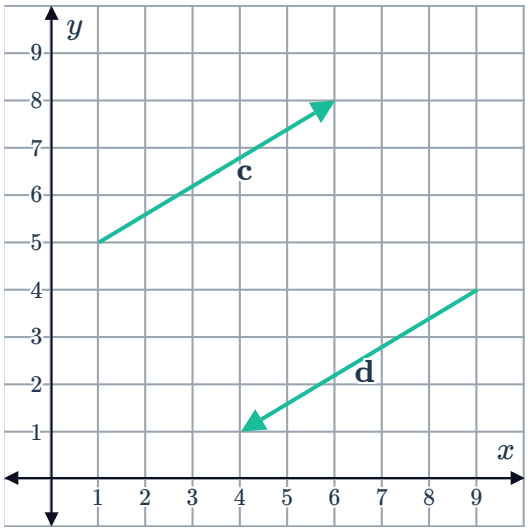
**i** Write the vector $c$ as a column vector.

**ii** Write the vector $d$ as a column vector.

**a**




**b**




**4** If $A = (-2, 3)$, $B = (1, 7)$, $C = (-9, 6)$ and $O$ is the origin, express each of the following as a column vector:

**a** $\overrightarrow{OB}$ **b** $\overrightarrow{AC}$ **c** $\overrightarrow{AO}$ **d** $\overrightarrow{CB}$

**5** Plot the vector $\begin{bmatrix} 5 \\ 9 \end{bmatrix}$ on a number plane. Use the origin as the starting point of the vector.

**6** Let $A = (-6, 2)$, $B = (1, 4)$, $C = (-3, 5)$ and $O$ be the origin. Plot the following vectors on a number plane:

**a** $\overrightarrow{OC}$ **b** $\overrightarrow{AB}$ **c** $\overrightarrow{BO}$ **d** $\overrightarrow{OC}$

**7** For each of the following sets of points:

**i** Plot the vectors $\overrightarrow{AB}$ and $\overrightarrow{CD}$ on on a number plane.

**ii** State whether the two vectors are equivalent.

**a** $A\,(5, 4)$, $B\,(5, 9)$, $C\,(-4, -2)$ and $D\,(-4, 3)$.

**b** $A\,(-4, 2)$, $B\,(-6, 6)$, $C\,(-4, -2)$ and $D\,(-6, 2)$.

**c** $A\,(3, -4)$, $B\,(3, -6)$, $C\,(5, 3)$ and $D\,(5, 5)$.

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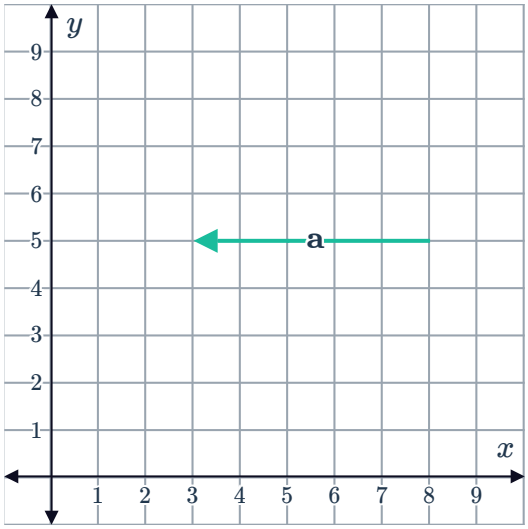
**8** State whether each of the following is a vector quantity:

- **a** A force of $8\ \mathrm{N}$ acting horizontally right.
- **b** A displacement of $4\ \mathrm{m}$ along the line joining $A$ and $B$.
- **c** A mass of $1\ \mathrm{kg}$
- **d** A time of $1$ second.

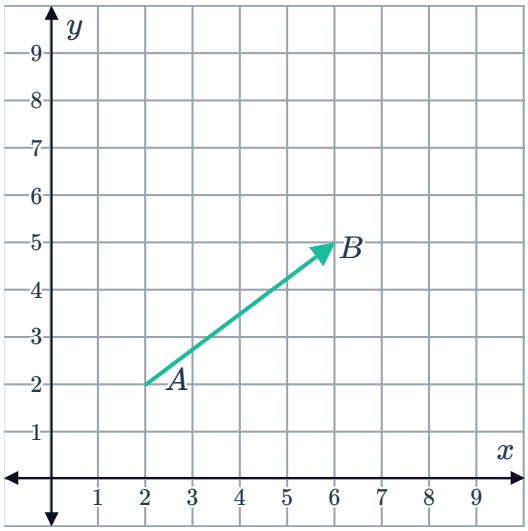
**9** What is the opposite of the vector representing $800$ km south?

## Magnitude of a vector (Extended)

**10** Find the magnitude of vector $a$:

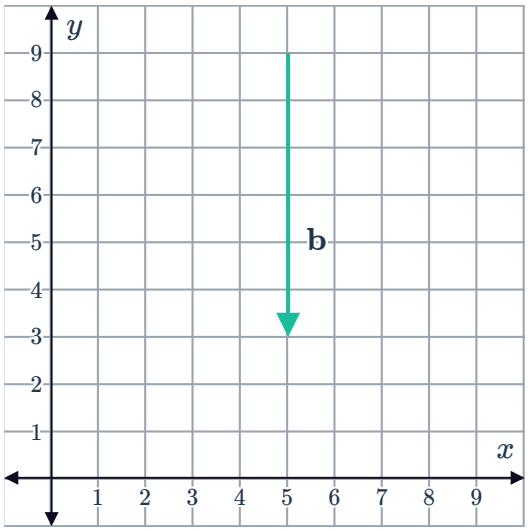



**11** Find the magnitude of the vector between points $A$ and $B$:

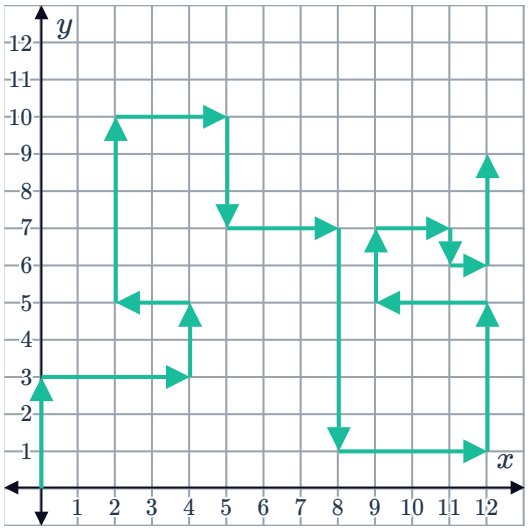



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**12** Find the magnitude of vector $b$:




**13** Consider the plotted path:




**a** Calculate the displacement of the path.

**b** Calculate the distance of the path.

**c** Is the displacement equal to the distance? Explain your answer.

**14** Consider the vector defined by the directed line segment from $(-5, 3)$ to $(1, -5)$.

**a** Plot the vector.

**b** Find the magnitude of the vector.

**15** Two vectors that are parallel can have different magnitudes. Is it true or false?

**16** Find the exact magnitudes of the following vectors

**a** $\begin{bmatrix} -7 \\ 0 \end{bmatrix}$ **b** $\begin{bmatrix} 5 \\ -7 \end{bmatrix}$ **c** $\begin{bmatrix} 8 \\ 15 \end{bmatrix}$ **d** $\begin{bmatrix} -8 \\ -15 \end{bmatrix}$

**e** $\begin{bmatrix} 6 \\ 0 \end{bmatrix}$ **f** $\begin{bmatrix} -6 \\ 0 \end{bmatrix}$

**17** Find the exact magnitude of the vector $(4, 8)$.

**18** Find the magnitude of the vector $(12, -4, 3)$.

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**19** If $a = (3x, 4x)$ and $|a| = 15$, find the value of $x$.

**20** A pilot flies $12$ km north and then $16$ km east. Find the magnitude of the vector created from her initial position to the final position.

**21** A boat travels $12$ km west and then $5$ km south. Determine the distance of the boat from its initial position.

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